

Mara Kirdani-Ryan

✉ marakr@uw.edu | 🌐 www.marakr.com

I work to move computing & information from forces that magnify hegemonic oppression towards forces that amplify relationality, right relation, and anticolonial pedagogies. I draw from sociology, social work, learning sciences, somatic education, and Indigenous knowledge framings to pursue organizational and collective change.

Education

University of Washington

Seattle, WA

Ph.D. in Computer Science & Engineering

2018 – 2024

Dissertation: Identity Fragmentation in Post-Secondary Computing Education

Amy J. Ko (Chair), Jennifer Mankoff, R. Ben Shapiro, Jondou Chen (GSR)

Carnegie Mellon University

Pittsburgh, PA

B.S. & M.S. in Electrical & Computer Engineering

2013 – 2017

Professional Appointments

University of Washington • The Information School

Seattle, WA

Assistant Teaching Professor

2024 - Present

- '25-'26: Cooperative Software Development (INFO 442), Intellectual Foundations of Informatics (INFO 200), and Gender & Information Technology (INFO 102)
- '24-'25: Cooperative Software Development (INFO 442), Client-Side Development (INFO 340), and Gender & Information Technology (INFO 102)

University of Washington • Code & Cognition Lab

Seattle, WA

Doctoral Researcher • Advised by Amy Ko

2018-2024

- Researching cultural change within computing; surfacing computing's cultural norms and challenging address them

Carnegie Mellon University • Abstract Research Group

Pittsburgh, PA

Research Assistant • Advised by Brandon Lucia

2017 – 2018

- Explored intermittent HW design and extended software models to guarantee consistency across HW interrupts

Apple Computer

Cupertino, CA

GPU Software Intern

May 2017 – August 2017

- Transitioned memory architecture for Intel GPU driver to map-based system

Cisco Systems • Cloud Platform Team

Boxborough, MA

Software Engineering Intern

May 2016 – August 2016

- Worked with application teams to implement infrastructure for alert correlating and squashing

National Security Agency

Baltimore, MD

CAE in Cyber Operations Intern

June 2015 – August 2015

- Surveyed a variety of security contexts and threat models and worked on a challenge problem of high operational interest; TS/SI clearance

University of Rochester • Music Research Lab

Rochester, NY

NSF REU Scholar • Advised by Mark Bocko

June - August, 2014

- Designed and created an acoustic camera, programmed in Matlab

Teaching Scholarship

University of Washington • The Information School

Seattle, WA

Critical Information Research Cohort

2026 - Ongoing

- An experiment towards relational pedagogy that situates existing Informatics Curriculum in a year-long cohort
- In collaboration with Nassim Parvin & Jason Young

AI Teaching Fellows Program

2026 - Ongoing

- Enacted a mentoring program to aid faculty in adjusting pedagogy in response to AI, addressing persistent issues in postsecondary education
- In collaboration with Joel Ross & Belén Salidas

- Aims to encourage a relational toolkit earlier in the informatics curriculum where students develop their own practices of collaboration & negotiation that can be iterated & fine-tuned throughout their time in the program
- In collaboration with Wanda Pratt, Heather Whiteman, and Lucas Moe

Cooperative Software Development Redesign

2025 - Ongoing

- Reframed existing practices around collaboration, software, and project management to center relationality & right relationship, creating a learning ecology that centers students' revision of their own relational toolkit
- Contextualized in four strata: relation with self, with collaborators, with community, and with Land, to encourage generalizable "soft" skills that transfer throughout students' professional & personal context

Publications

Peer-Reviewed

Narratives on Cheating in Computing Education

Mara Kirdani-Ryan, Jayne Everson, Eman Sharif, Amy J Ko

ACM Transactions on Computing Education (TOCE). ACM, In Review

Neurodivergent Legitimacy in Computing Spaces

Mara Kirdani-Ryan, Amy J Ko

ACM Transactions on Computing Education (TOCE) pp. 1–28. ACM, 2024

Exploring the Impact of Assessment Policies on Marginalized Students' Experiences in Post-Secondary Programming Courses

Eman Sherif, Jayne Everson, F. Megumi Kivuva, Mara Kirdani-Ryan, Amy J. Ko

Proceedings of the 2024 ACM Conference on International Computing Education Research-Volume 1, 2024

"Taught to be Automata": Examining the Departmental Role in Shaping Initial Career Choices of Computing Students

Mara Kirdani-Ryan, Amy J Ko, Emilia A Borisova

Computer Science Education pp. 1–27. Taylor & Francis, 2023

The House of Computing: Integrating Counternarratives into Computer Systems Education

Mara Kirdani-Ryan, Amy J Ko

Proceedings of the 53rd ACM Technical Symposium on Computer Science Education pp. 279–285. ACM, 2022

Book Chapters

Operating Systems

Amy J. Ko, Mara Kirdani-Ryan

Chapter in: Critically Conscious Computing: Methods for Secondary Education, <https://criticallyconsciouscomputing.org/>

Edited by: Amy J. Ko, Anne Beitlers, Brett Wortzman, Matt Davidson, Alannah Oleson, Mara Kirdani-Ryan, Stefania Druga, and Jayne Everson, 2023

Magazine Articles

It's time for more Critical CS Education

Amy J Ko, Alannah Oleson, Mara Kirdani-Ryan, Yim Register, Benjamin Xie, Mina Tari, Matthew Davidson, Stefania Druga, Dastyni Loksa

Communications of the ACM pp. 31–33. ACM, 2020

Teaching Experience

University of Washington

Seattle, WA

Instructor of Record · Cooperative Software Development (9 sections)

2024-2026

Instructor of Record · Becoming an Information Professional

Autumn 2026

Instructor of Record · Client Side Development (2 sections)

2025 - 2026

Instructor of Record · Intellectual Foundations of Informatics

Winter 2026

Instructor of Record · Gender & Information Technology (2 sections)

2025 - 2026

Instructor of Record · Deconstructing Cultural Norms in Computer Science

Spring 2023

Teaching Assistant · The Hardware/Software Interface

Winter 2023

Teaching Assistant · The Hardware/Software Interface

Winter 2022

Instructor of Record · The Hardware/Software Interface

Summer 2021

Teaching Assistant · The Hardware/Software Interface

Spring 2021

Teaching Assistant · User Interface Software and Technology

Winter 2021

Carnegie Mellon University*Pittsburgh, PA*

Head Teaching Assistant · Introduction to Embedded Systems

Spring 2018

Teaching Assistant · Secure Coding

Spring 2018

Teaching Assistant · Introduction to Embedded Systems

Fall 2017

Teaching Assistant · Introduction to Computer Architecture

Spring 2017

Teaching Assistant · Logic Design and Verification

Fall 2016

Teaching Assistant · Structure and Design of Digital Systems

Spring 2016

Teaching Assistant · Structure and Design of Digital Systems

Fall 2015

Teaching Assistant · Structure and Design of Digital Systems

Spring 2015

Awards & Honors

2018 **Dora Zee Ling Endowed Fellowship in Computer Science & Engineering***University of Washington*2017 **ECE Department Outstanding Teaching Assistant***Carnegie Mellon University*

Invited Panels

2023 **Huskies for Neurodivergence Panel***University of Washington*2023 **Neurodiverse Students in STEM CBI Panel***University of Washington*

Birds of a Feather (Organizer)

2024 **Cultivating and Celebrating LGBTQ+ Community in Computing Education***SIGCSE*2024 **Finding Neurodivergent Community in Computing Education***SIGCSE*2023 **Finding Neurodivergent Community in Computing Education***SIGCSE*